

# Human Respiratory and Excretion System

## Lecture 20

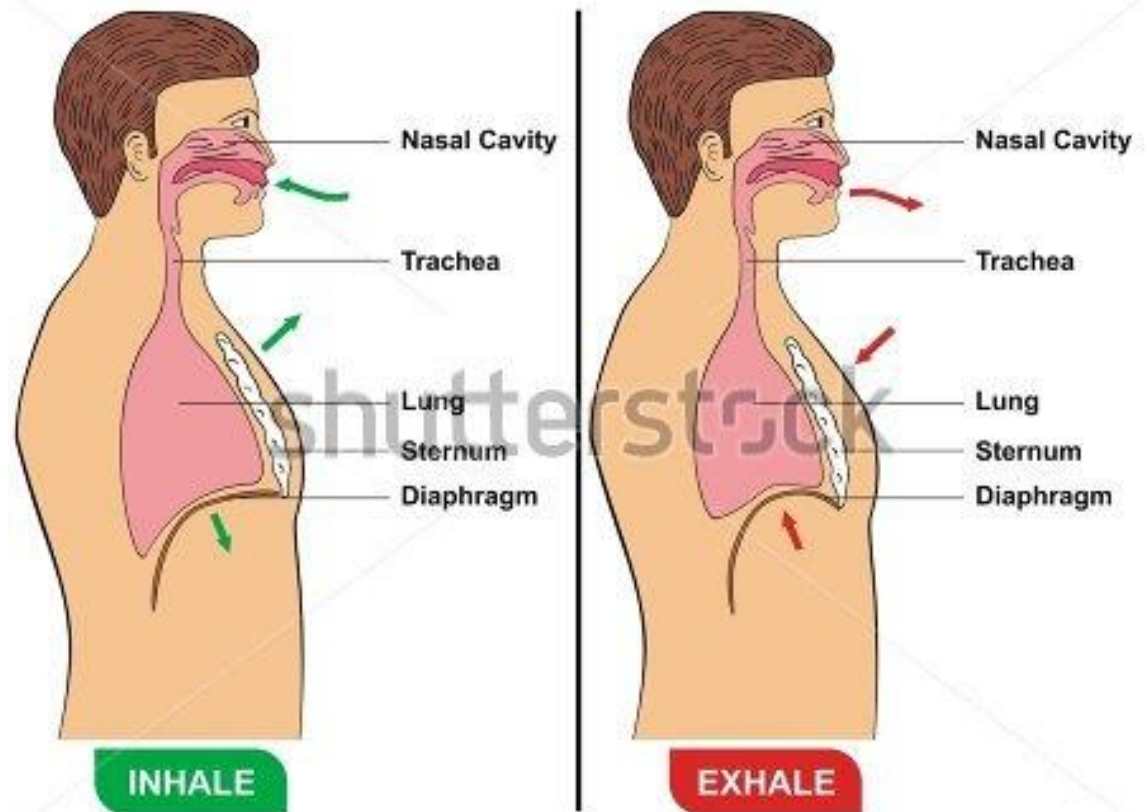
**In 24 hours:**

$6 \text{ L/min} \times 60 \text{ min} \times 24 \text{ h} \approx \textbf{8,640 L/day}$  ( $\sim 8.6 \text{ m}^3$  of air).

During exercise, this can rise **5–10×**.

# Gas Exchange and Breathing

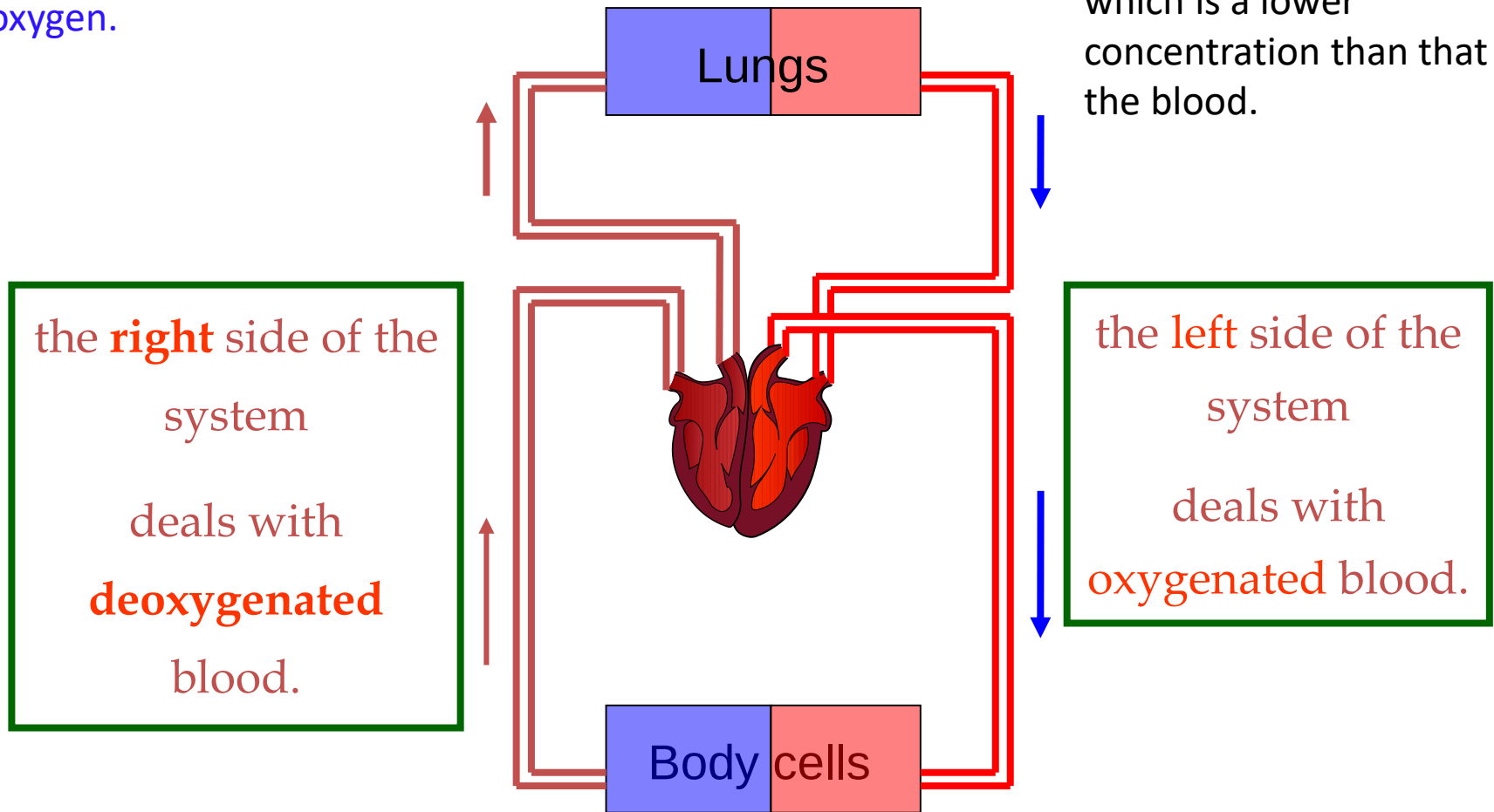
- Aerobic animals → → **need oxygen,  $O_2$**  for energy production (**Cellular Respiration**)
- **Carbon dioxide,  $CO_2$**  → too much of it can be **harmful**
- The *respiratory* and *circulatory* systems **work together** to **deliver oxygen** to cells and **remove carbon dioxide** in a two-phase process called **respiration**.
- Two phases:
  - **Inhalation**
  - **Exhalation**



# Gas Exchange in Lungs

Fresh inhaled air in the airspace contains about 21% oxygen.

Fresh air contains only 0.034% carbon dioxide, which is a lower concentration than that in the blood.



# The Human Respiratory or Gas Exchange System

- **Nose to the lungs**
- The *upper respiratory tract* consists of
  - (1) the *nose* or nasal passages and
  - (2) the *pharynx*, or throat.
- The *lower respiratory tract* includes
  - (3) the *larynx*, or voice box;
  - (4) the *trachea*, or windpipe, which splits into two main branches called *bronchi*; tiny branches of the **bronchi** called *bronchioles*; and
  - (5) the *lungs*, a pair of saclike, spongy organs.

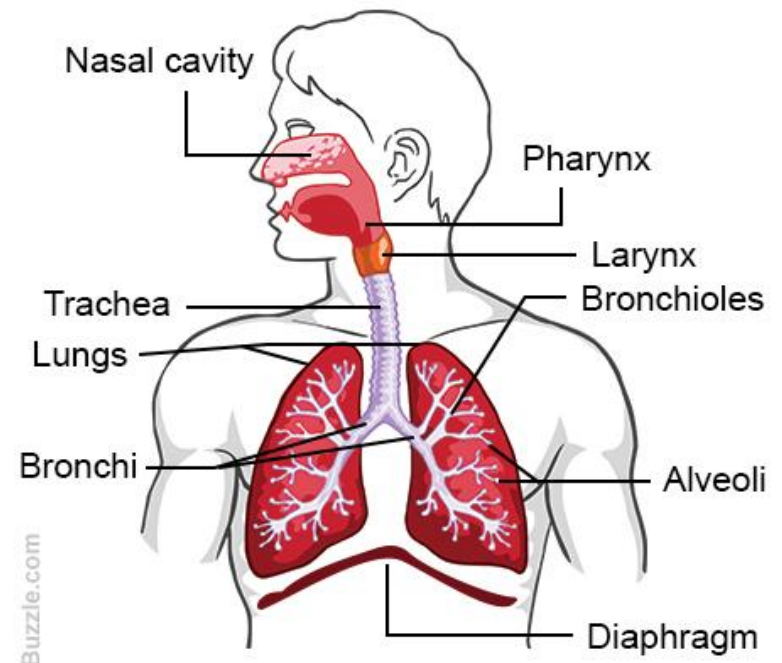
# Upper Respiratory Tract

## 1. Nasal Passages

The uppermost portion of the human respiratory system, the nose is a hollow air passage that functions in breathing and in the sense of smell. The nasal cavity moistens and warms incoming air, while small hairs and mucus filter out harmful particles and microorganisms.

## 2. Pharynx

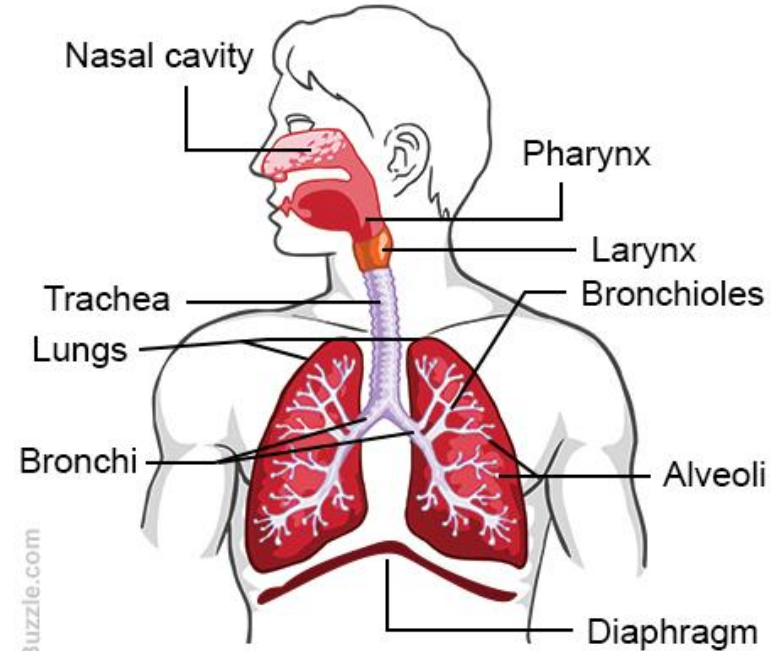
Air leaves the nasal passages and flows to the pharynx, a short, funnel-shaped tube about 13 cm (5 in) long that transports air to the larynx.



# Lower Respiratory Tract

## 3. Larynx

Air moves from the pharynx to the larynx, a structure about 5 cm (2 in) long located approximately in the middle of the neck.



## 4. Trachea, Bronchi, and Bronchioles

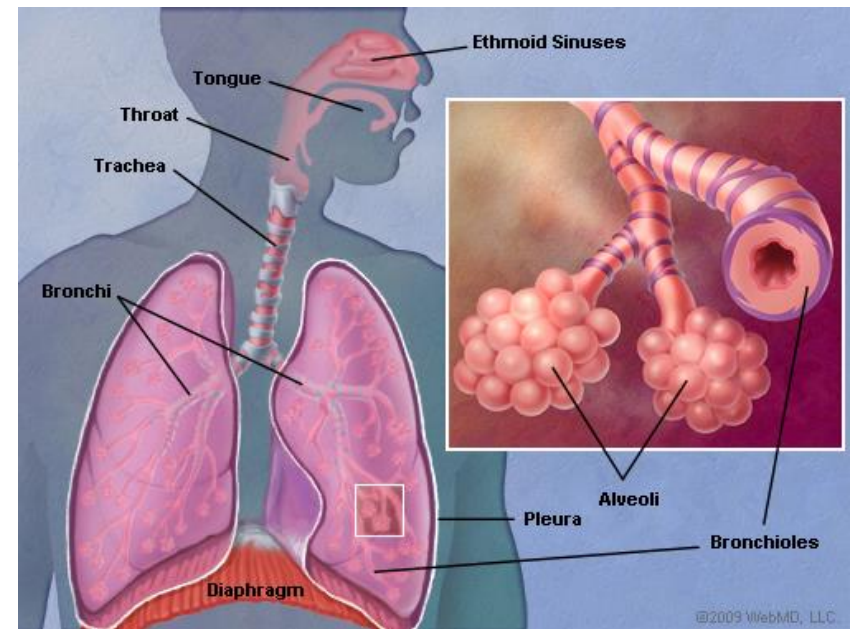
Air passes from the larynx into the trachea, a tube about 12 to 15 cm (about 5 to 6 in) long located just below the larynx. The trachea branches into two tubes, the *left and right bronchi*, which deliver air to the left and right lungs, respectively. Within the lungs, the bronchi branch into smaller tubes called *bronchioles*.

# Lower Respiratory Tract

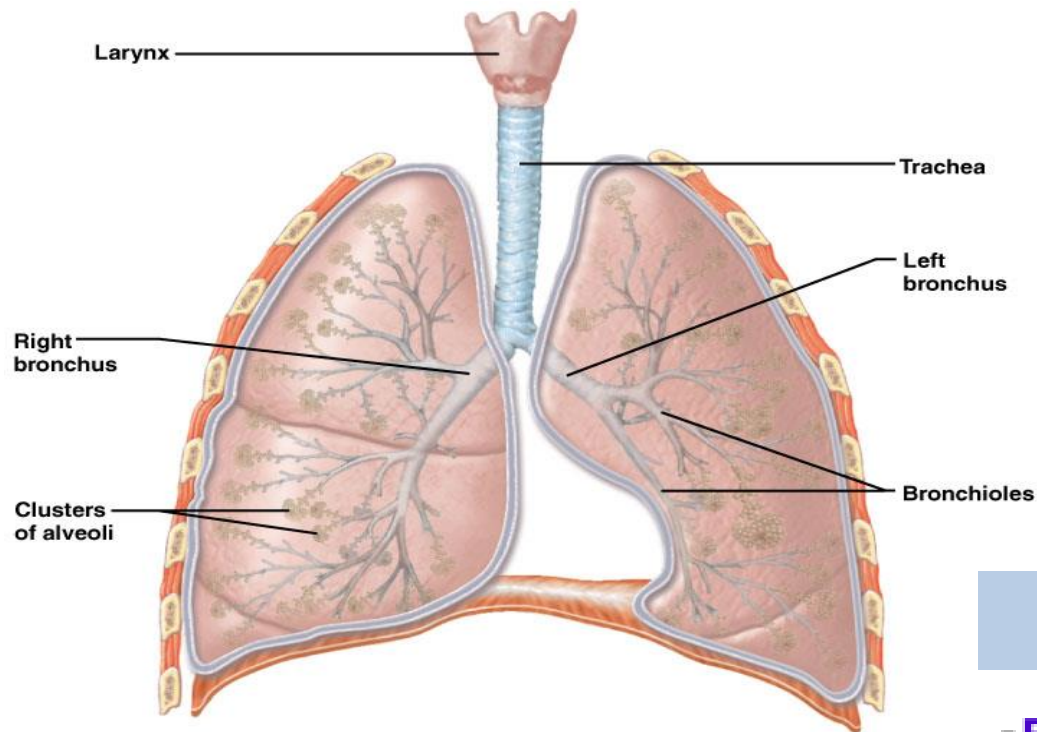
## 5. Lungs

The two branches of the trachea, called *bronchi*, subdivide within the lobes into smaller and smaller air vessels. They terminate in *alveoli*, tiny air sacs surrounded by capillaries.

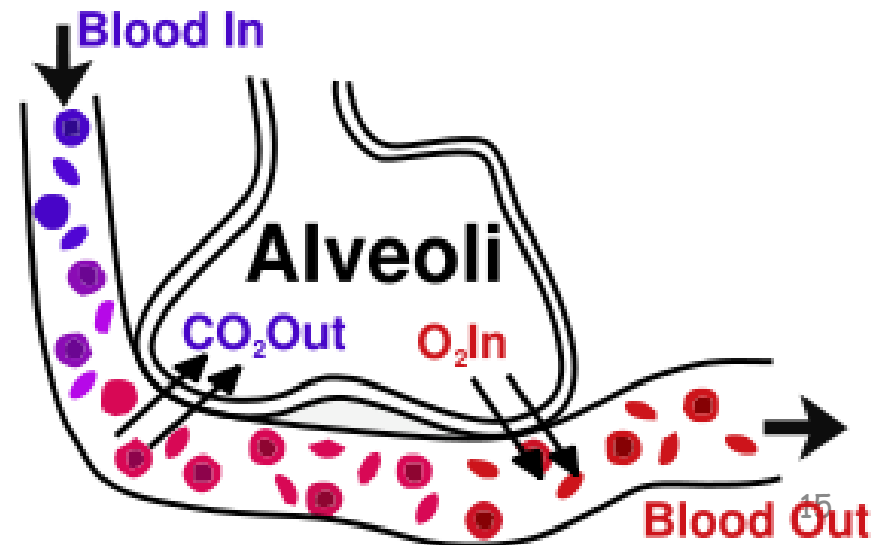
When the alveoli inflate with inhaled air, oxygen diffuses into the blood in the capillaries to be pumped by the heart to the tissues of the body, and carbon dioxide diffuses out of the blood into the lungs, where it is exhaled.



# Components of the Lower Respiratory Tract



## Gas Exchange in Lungs



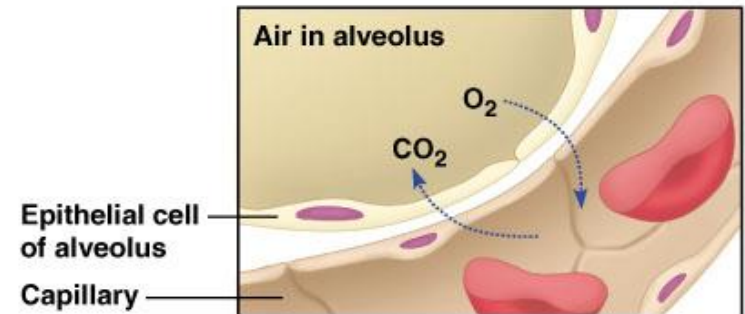
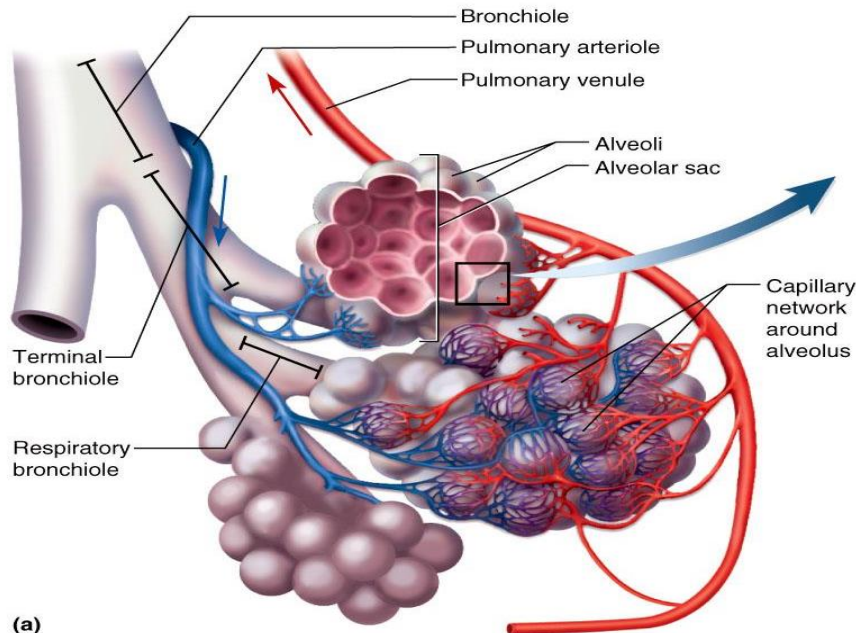
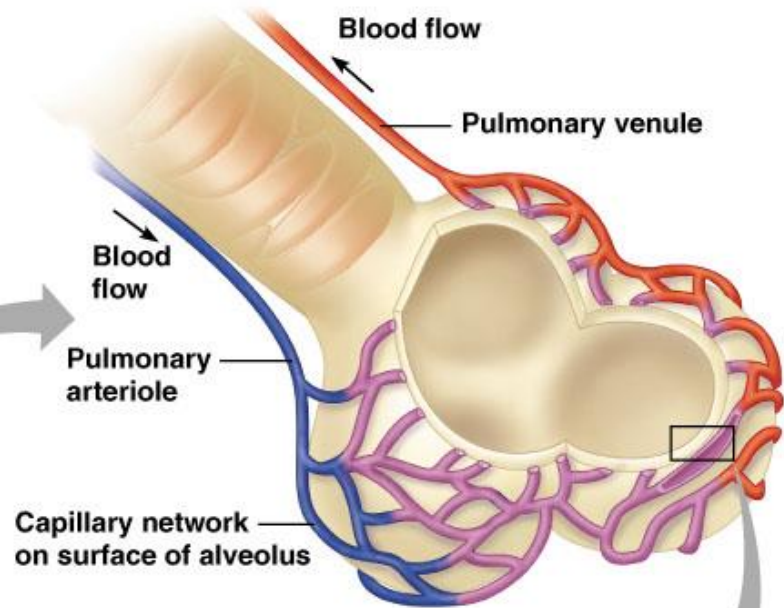
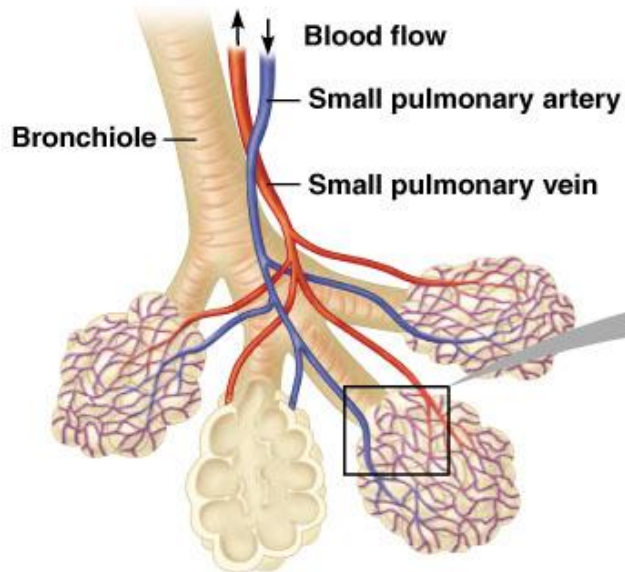
# Gas Exchange in Lungs

## Composition of Breath

| Components of Earth's Atmosphere | Composition of Air Breathed <b>IN</b> (%) | Composition of Air Breathed <b>OUT</b> (%) |
|----------------------------------|-------------------------------------------|--------------------------------------------|
| Nitrogen                         | 78                                        | 75                                         |
| Oxygen                           | <b>21</b>                                 | 16                                         |
| Argon                            | 0.9                                       | 0.9                                        |
| Carbon Dioxide                   | 0.035                                     | <b>4.0</b>                                 |
| Water vapor                      | 0.4                                       | 4.0                                        |

Misconception alert! Carbon Dioxide and Oxygen ARE NOT the only things that make up the air we breathe!

# Gas Exchange Between the Blood and Alveoli



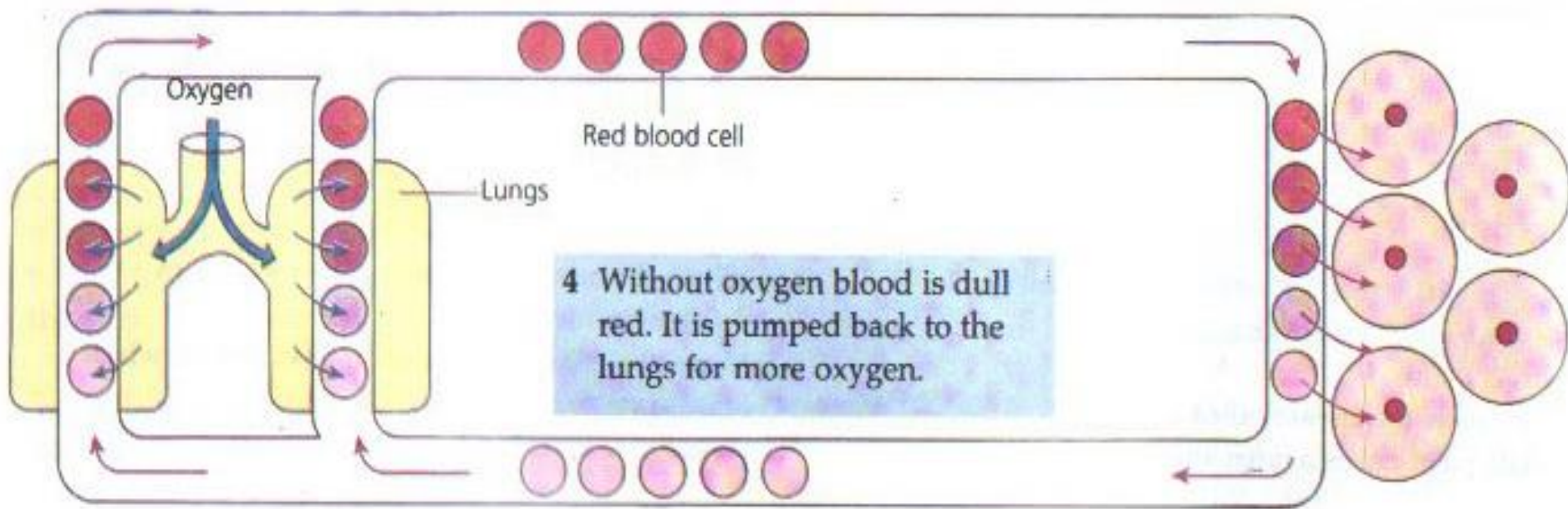
# How tissue gets oxygen and release carbon dioxide

## How Red blood Cells carry oxygen :

1 The red cells pick up oxygen as blood is pumped through the lungs.

2 The oxygen and haemoglobin join to form **oxyhaemoglobin**. This is bright red.

3 As the blood passes around the body, the oxyhaemoglobin breaks down and releases oxygen to the body cells.

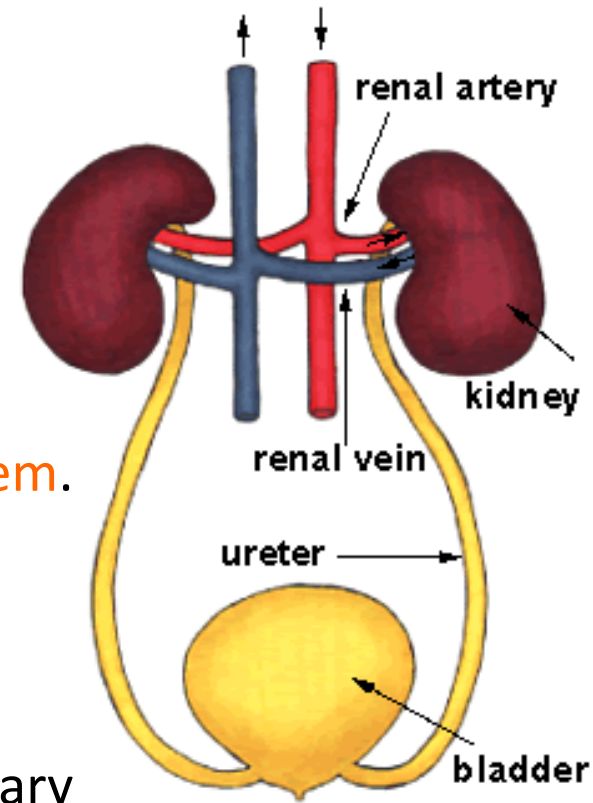


# THE EXCRETORY SYSTEM

- The excretory system helps to **remove body waste safely**.
- The kidneys act like **filters**.

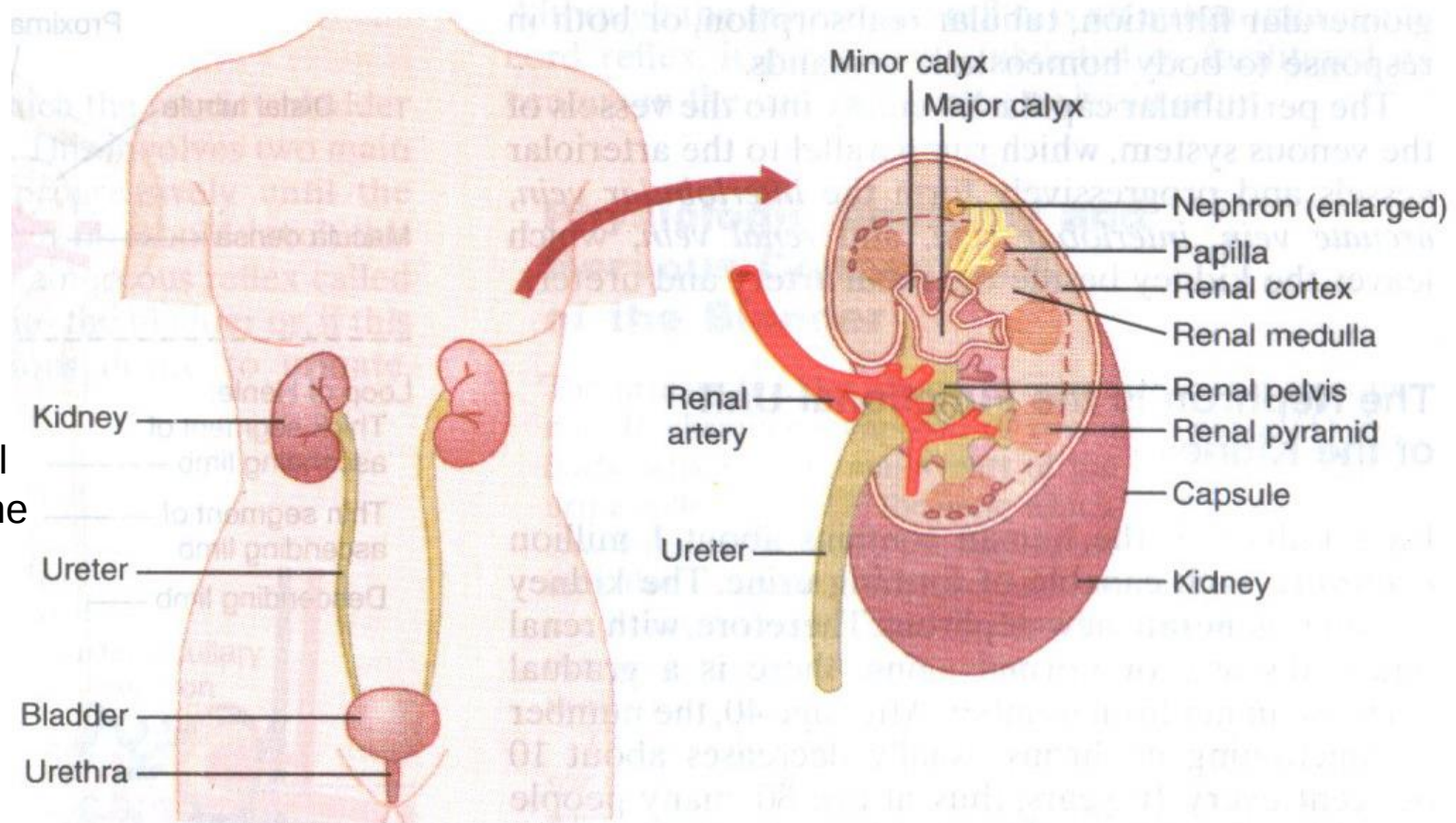
Kidneys are the **excretory organ** of our body. The followings are the **major components of urinary system**.

- **Urinary system** includes:
  - Kidneys**...formation of urine
  - Ureters**...transport of urine from kidneys to urinary bladder
  - Urinary bladder**...reservoir of urine
  - urethra**...passage of urine from urinary bladder to outside



# Kidney Structure

- Each kidney contains about one million **Nephrons** (filtering units) which are the structural & functional units of kidneys.



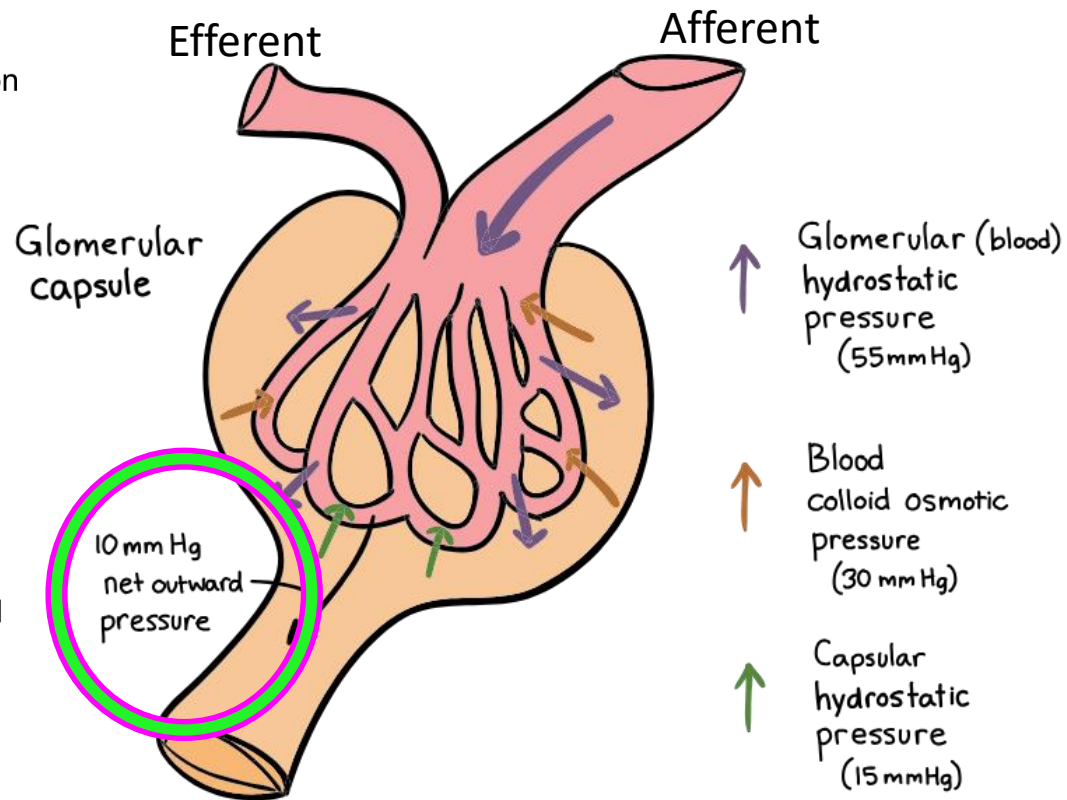
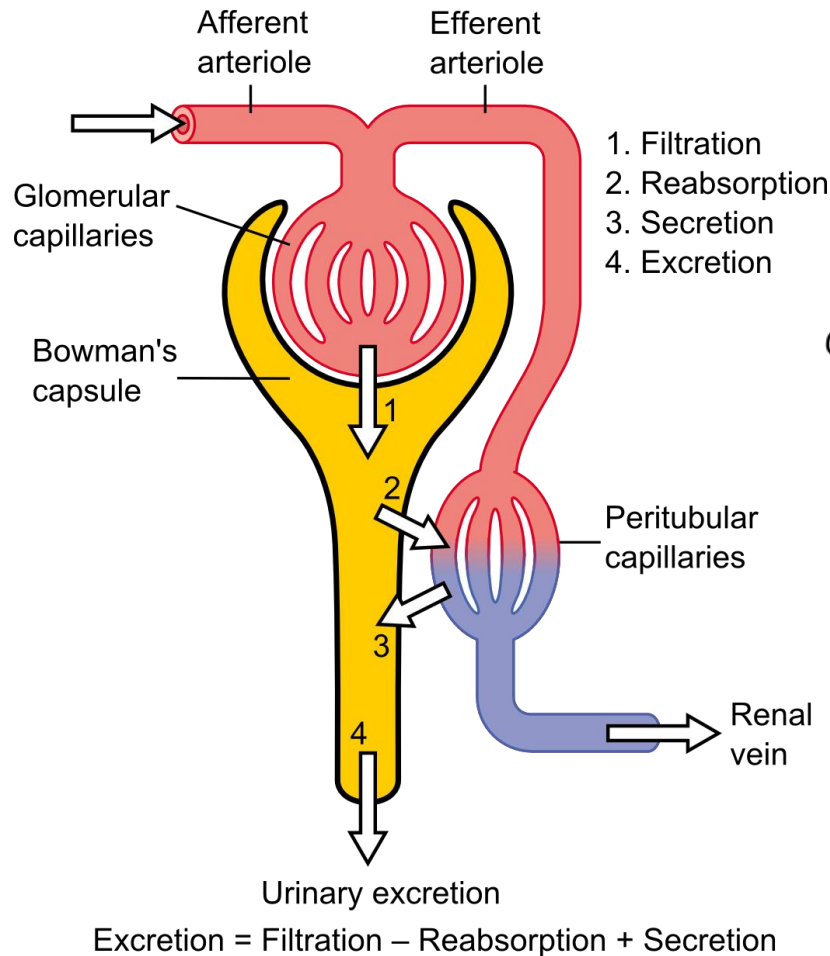
**Figure – General organization of the kidneys and the urinary system**

# Functions of kidneys

- Excretion of excess unnecessary substances & metabolic waste products
- Osmoregulation
- Regulation of -----:
  1. water & electrolyte balances
  2. acid-base balances
  3. blood pressure
  4. red cell production

*Osmoregulation is the process by which cells and simple organisms maintain fluid and electrolyte balance with their surroundings.*

# FILTRATION & REASBSORPTION

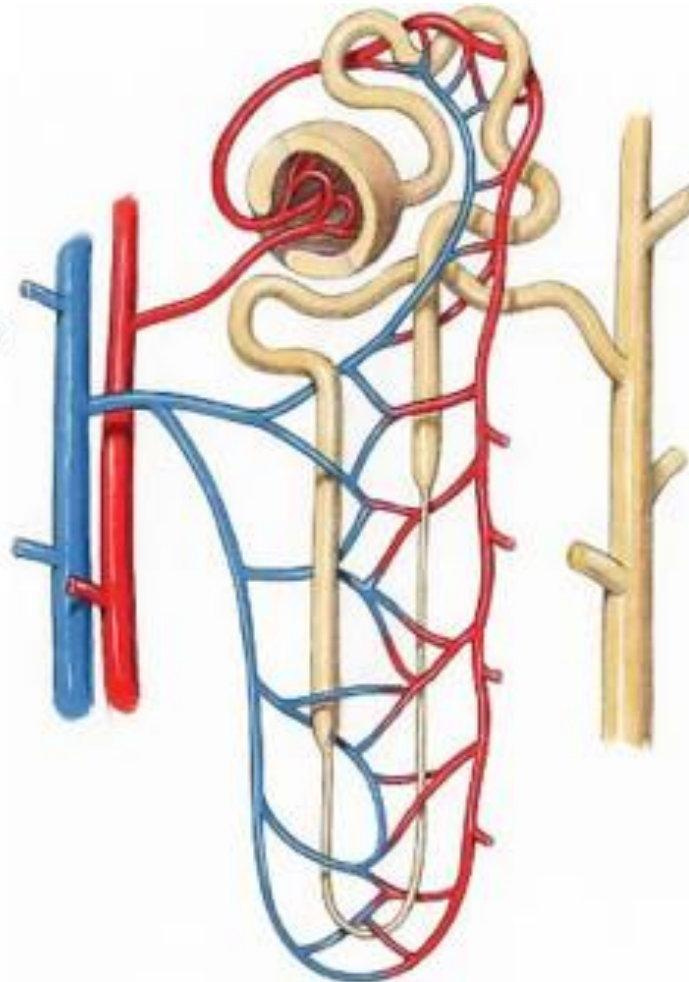


$$55 \text{ mm} - [45 \text{ mm} - (10 \text{ mm})] = +10 \text{ mm of Hg}$$

# Filtration & reabsorption

## Filtration

Most filtration occurs in the glomerulus. Blood pressure forces water, salt, glucose, amino acids, and urea into Bowman's capsule. Proteins and blood cells are too large to cross the membrane; they remain in the blood. The fluid that enters the renal tubules is called the filtrate.



## Reabsorption

As the filtrate flows through the renal tubule, most of the water and nutrients are reabsorbed into the blood. The concentrated fluid that remains is called urine.

# How kidneys clean blood

- Kidneys clean blood by filtering it. They filter all our **blood 300 times a day**. The filtering is done by over a million tubes packed into each kidney. These tubes are called **nephrons**.
- Some people's kidneys are not very good at filtering blood. Kidney machines help by filtering blood for them.

1,728 L/day of blood passes through both kidneys.

→ about **180 liters of filtrate/day** but Most of this is **reabsorbed**; only ~1.5–2 L becomes urine.